

The System Unit

What is the **system unit**?

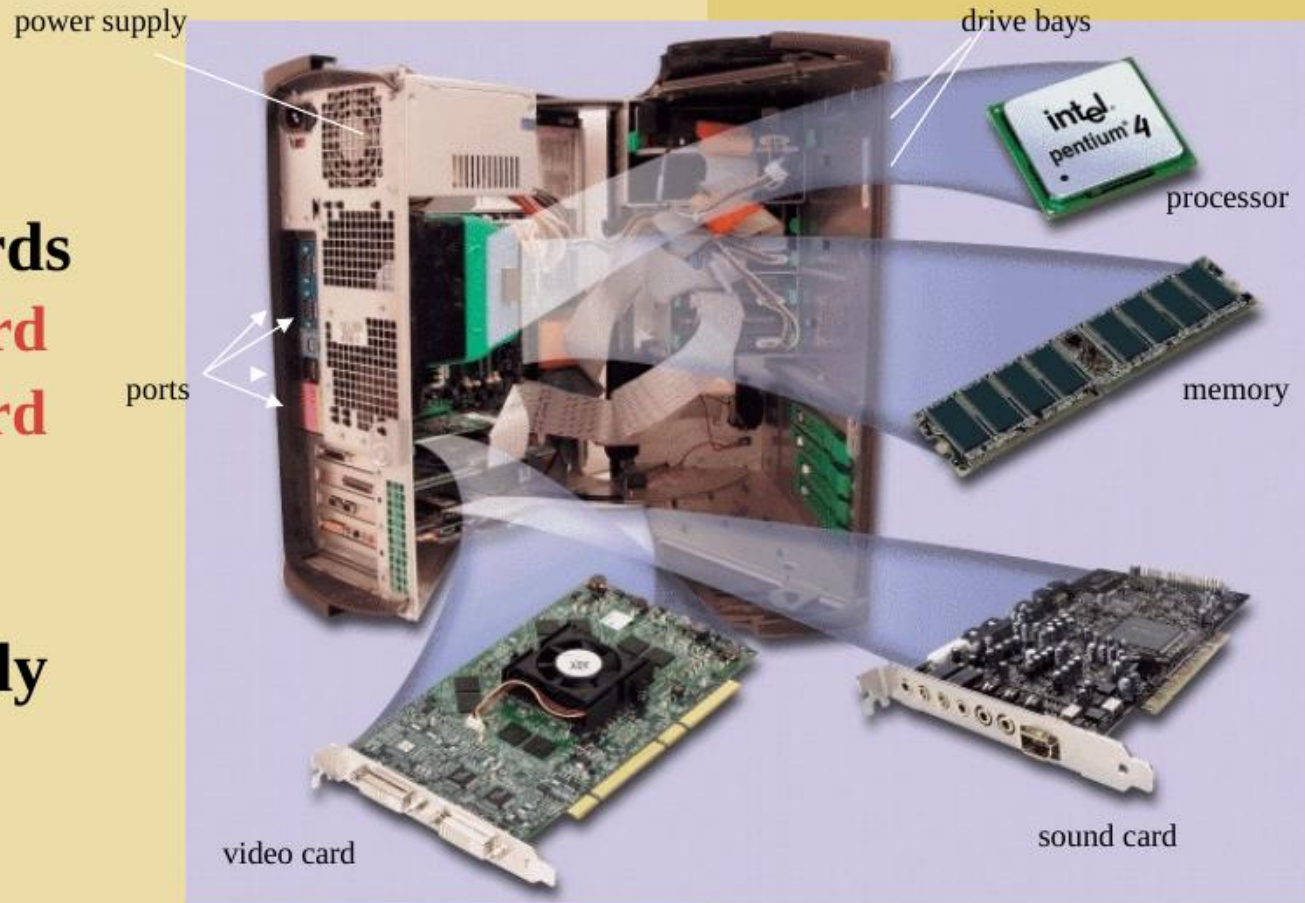
- Case that contains electronic components of the computer used to process data
- Sometimes called the chassis



The System Unit

What are common components inside the system unit?

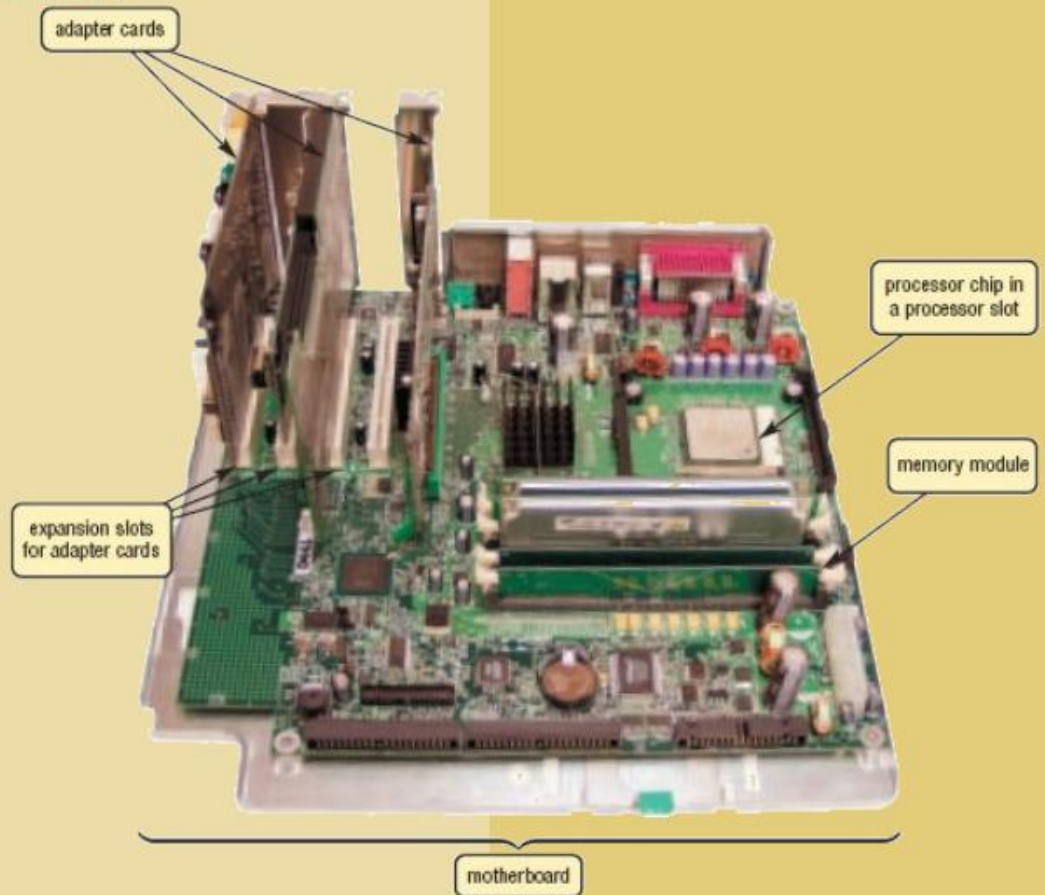
- **Processor**
- **Memory**
- **Adapter cards**
 - **Sound card**
 - **Video card**
- **Ports**
- **Drive bays**
- **Power supply**



The System Unit

What is the **motherboard**?

- Main circuit board in system unit
- Contains adapter cards, processor chips, and memory chips
- Also called system board



The System Unit

What is a **chip**?

- **Small piece of semi-conducting material on which integrated circuits are etched**
 - Integrated circuits contain many microscopic pathways capable of carrying electrical current
- **Chips are packaged so they can be attached to a circuit board**

Processor

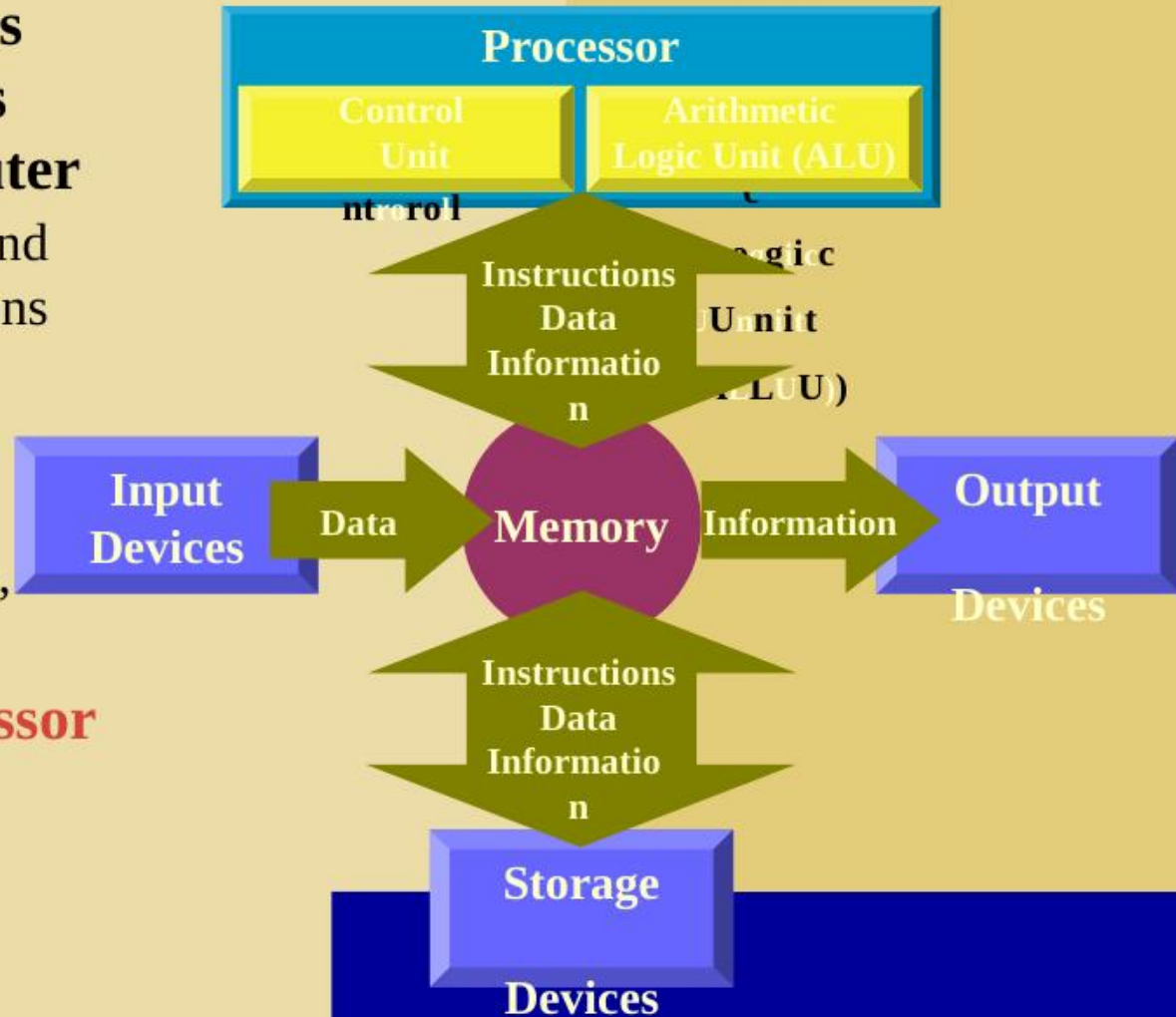
What is the **central processing unit (CPU)**?

- Interprets and carries out basic instructions that operate a computer

- **Control unit** directs and coordinates operations in computer

- **Arithmetic logic unit (ALU)** performs arithmetic, comparison, and logical operations

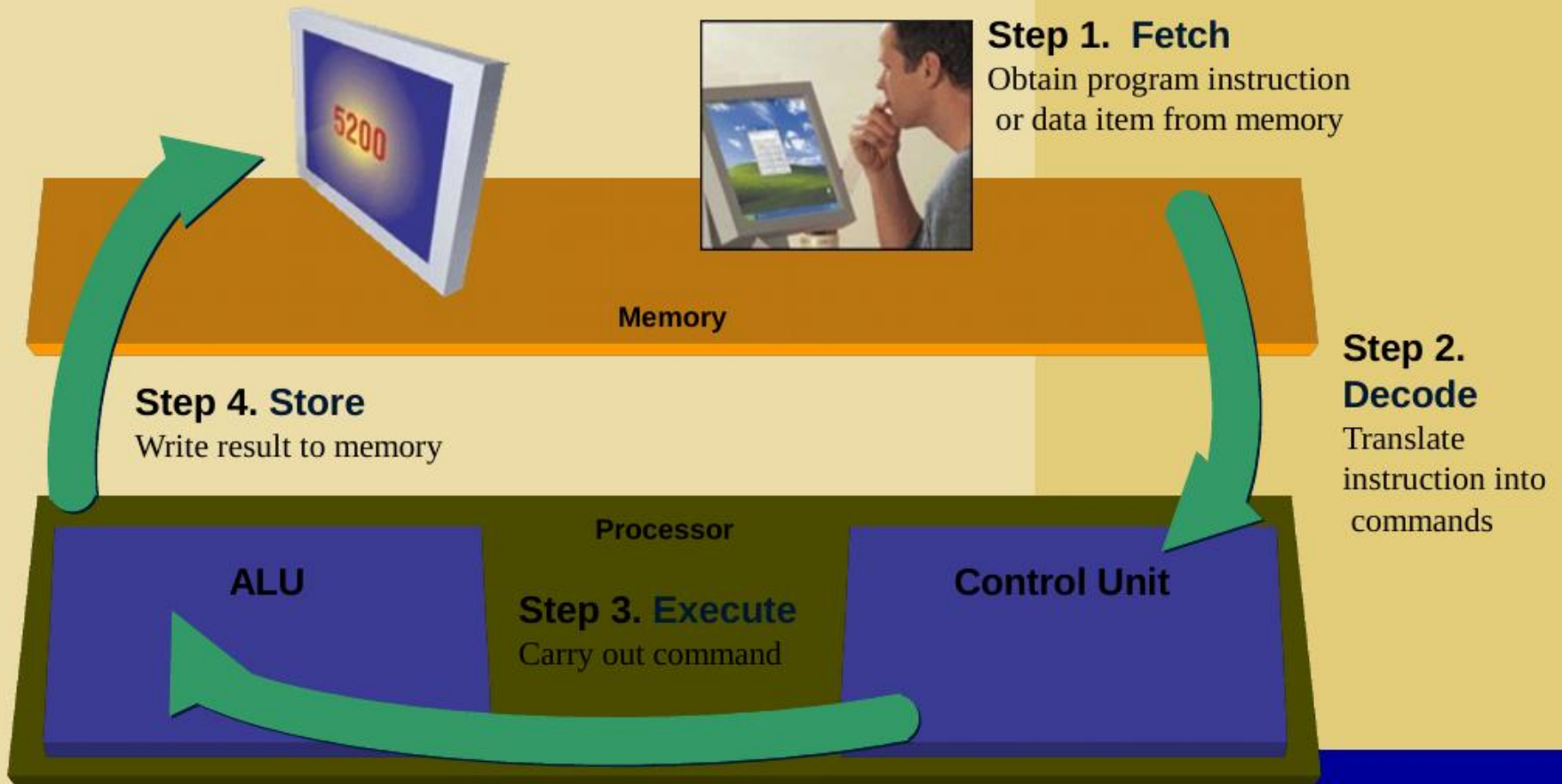
- Also called the **processor**



Processor

What is a machine cycle?

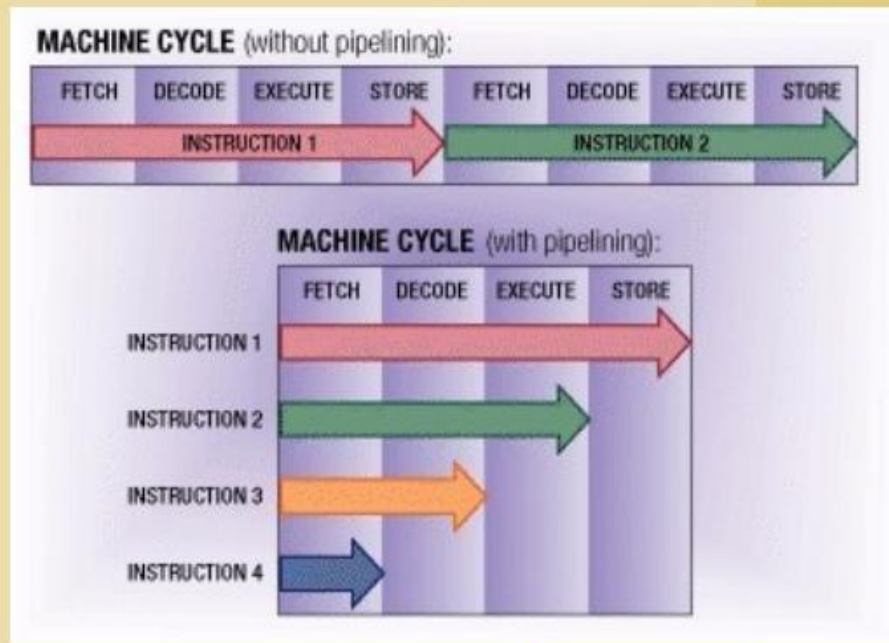
- Four operations of the CPU comprise a machine cycle



Processor

What is pipelining?

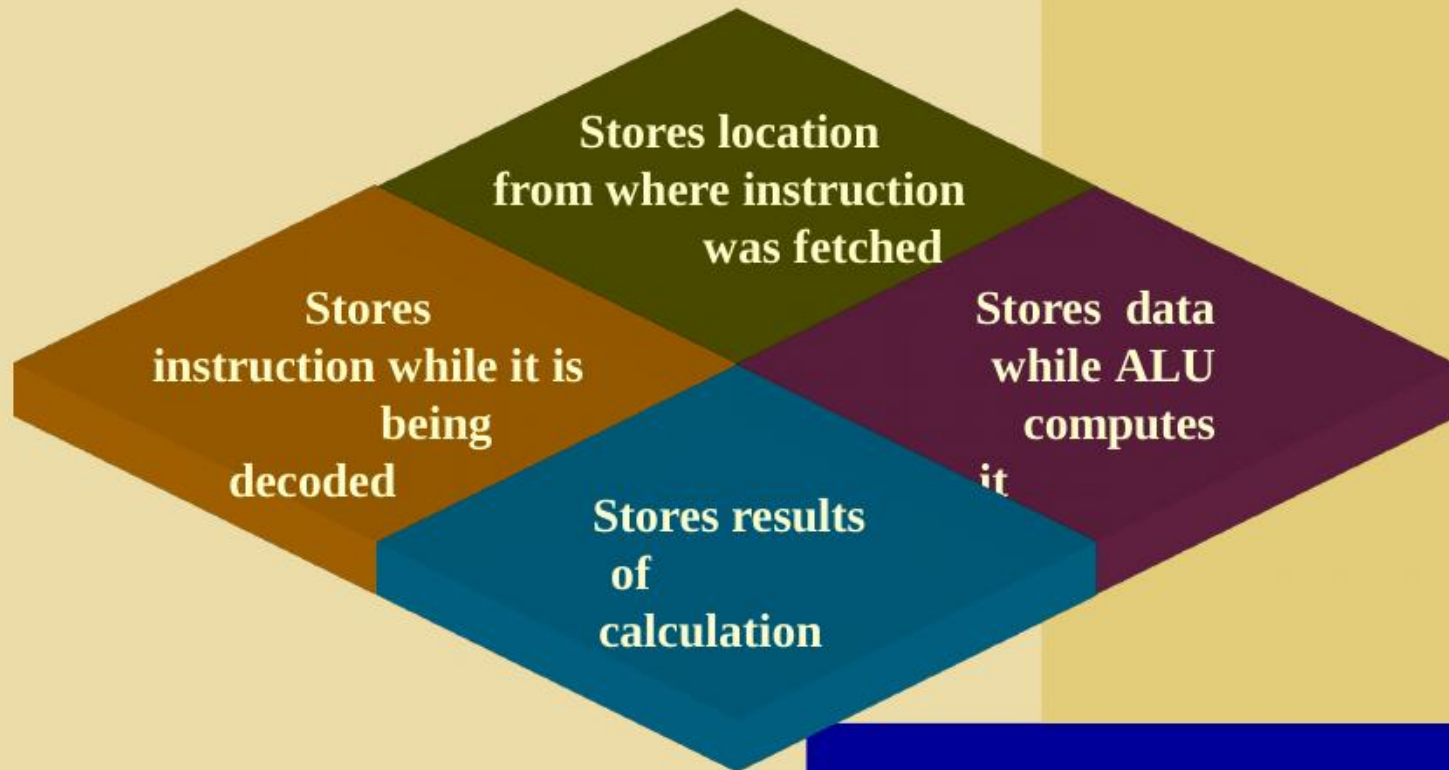
- CPU begins fetching second instruction before completing machine cycle for first instruction
- Results in faster processing



Processor

What is a register?

- Temporary high-speed storage area that holds data and instructions



Processor

What is the **system clock**?

- Controls timing of all computer operations
- Generates regular electronic pulses, or ticks, that set operating pace of components of system unit

Each tick
is a
**clock
cycle**

Pace of system
clock is **clock speed**
Most clock speeds are
in the gigahertz (GHz)
range (1 GHz = one
billion ticks of system
clock per second)

Processor speed can
also be measured in
**millions of instructions
per second (MIPS)**

Processor

Which processor should you select?

- The faster the processor, the more expensive the computer

Intel Processor	Desired Clock Speed
Itanium or Xeon 	1.3 GHz and up
Pentium family 	3.0 GHz and up 2.4 GHz to 3.0 GHz Up to 2.4 GHz
Celeron 	2.2 GHz and up

Processor

What are heat sinks, heat pipes, and liquid cooling?

- **Heat sink**—component with fins that cools processor
- **Heat pipe**—smaller device for notebook computers

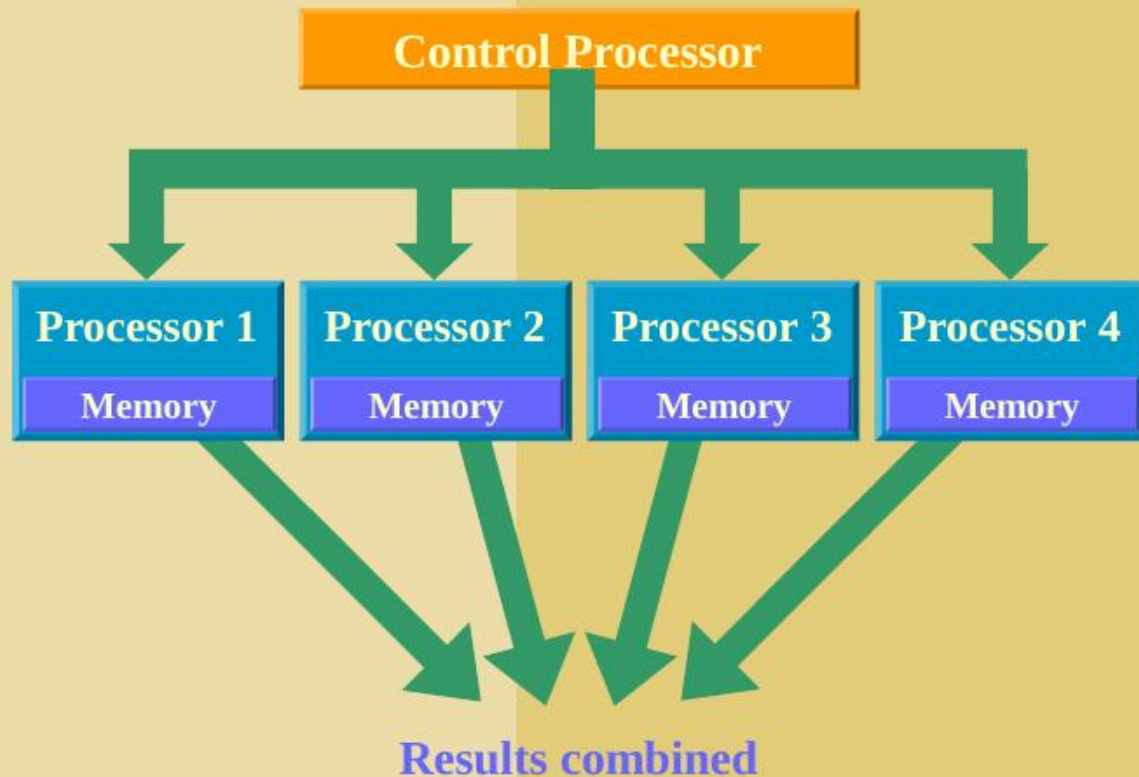
Liquid cooling—uses a continuous flow of fluids to transfer heat away



Processor

What is parallel processing?

- Using multiple processors simultaneously to execute a program faster
- Requires special software to divide problem and bring results together



Data Representation

How do computers represent data?

- Most computers are **digital**

BINARY DIGIT (BIT)	ELECTRONIC CHARGE	ELECTRONIC STATE
1		ON
0		OFF

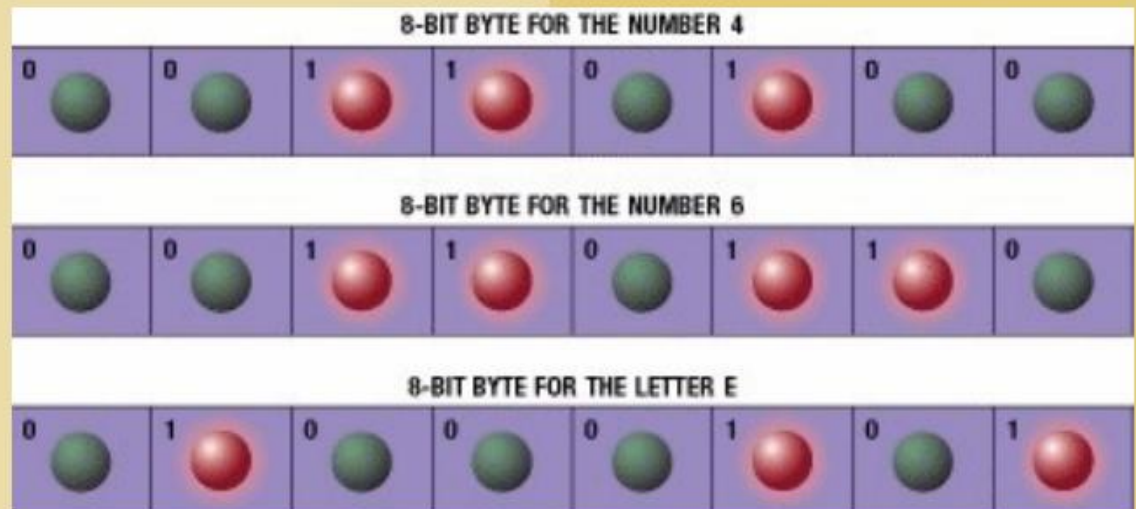
- Recognize only two discrete states: on or off
- Use a **binary system** to recognize two states
- Use Number system with two unique digits: 0 and 1, called **bits** (short for binary digits)

Data Representation

What is a **byte**?

- Eight bits grouped together as a unit
- Provides enough different combinations of 0s and 1s to represent 256 individual characters

- Numbers
- Uppercase and lowercase letters
- Punctuation marks
- Other



Data Representation

What are three popular coding systems to represent data?

- **ASCII—American Standard Code for Information Interchange**
- **EBCDIC—Extended Binary Coded Decimal Interchange Code**
- **Unicode—coding scheme capable of representing all world's languages**

ASCII	Symbol	EBCDIC
00110000	0	11110000
00110001	1	11110001
00110010	2	11110010
00110011	3	11110011

Data Representation

How is a letter converted to binary form and back?



Step 1.

The user presses the capital letter **D** (shift+D key) on the keyboard.



Step 4.

After processing, the binary code for the capital letter **D** is converted to an image, and displayed on the output device.



Step 3.

The signal for the capital letter **D** is converted to its ASCII binary code (01000100) and is stored in memory for processing.

Step 2.

An electronic signal for the capital letter **D** is sent to the system unit.

Memor

y

What is **memory**?

- Electronic components that store instructions, data, and results
- Consists of one or more chips on motherboard or other circuit board
- Each byte stored in unique location called an address, similar to seats in a concert hall



Memor

y

How is memory measured?

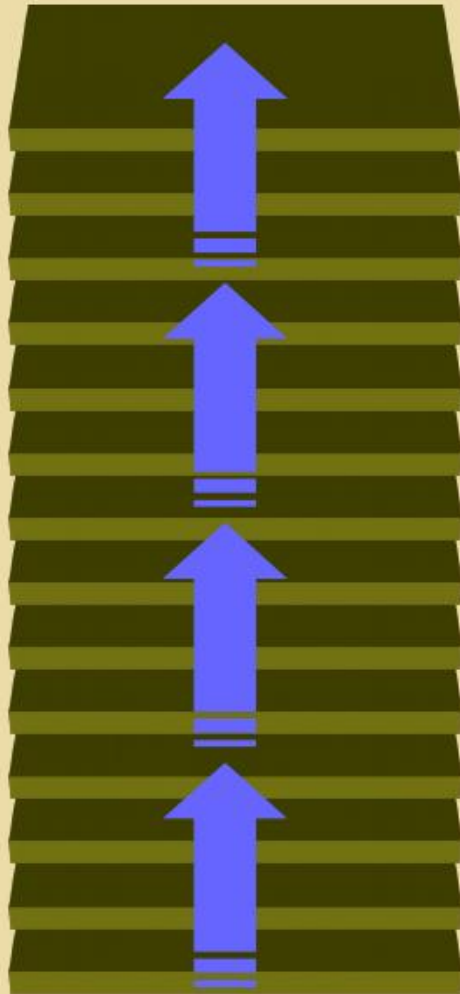
- **By number of bytes available for storage**

Term	Abbreviation	Approximate Size
Kilobyte	KB or K	1 thousand bytes
Megabyte	MB	1 million bytes
Gigabyte	GB	1 billion bytes
Terabyte	TB	1 trillion bytes

Memor

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What is random access memory (**RAM**)?



Memory chips that can be read from and written to by processor

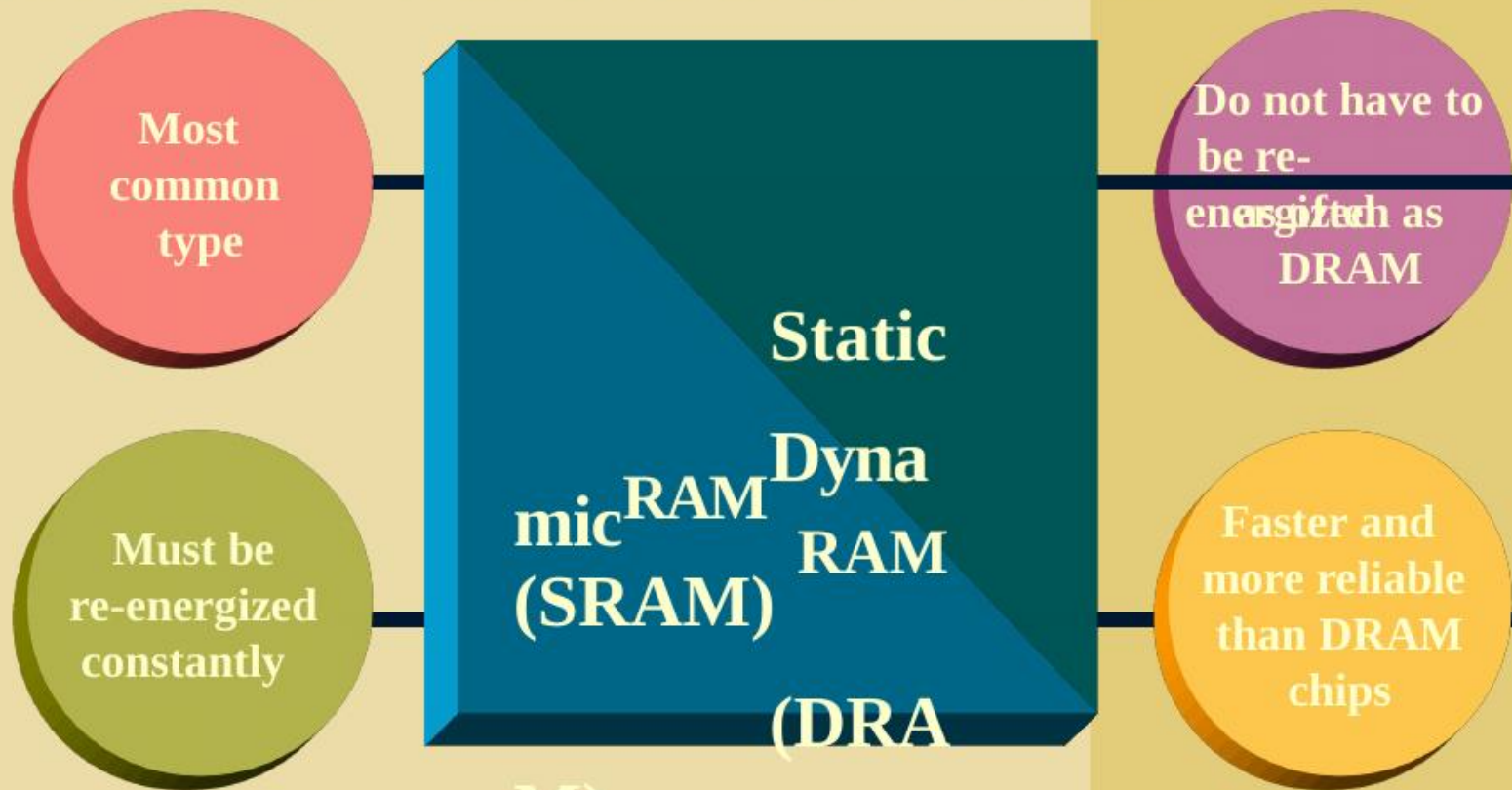
Also called main memory or primary storage

Most RAM is **volatile**, it is lost when computer's power is turned off

The more RAM a computer has, the faster it responds

Memory

y
What are two basic types of RAM chips?



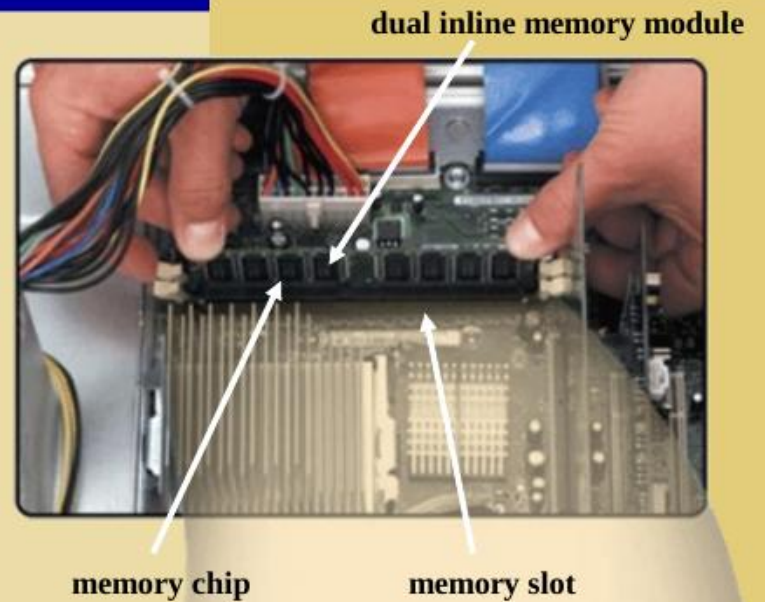
Newer Type: Magnetoresistive RAM (MRAM)

Memor

y

Where does memory reside?

- Resides on small circuit board called **memory module**
- **Memory slots** on motherboard hold memory modules



Memor

y

How much RAM does an application require?

- **Depends on the types of software you plan to use**
- **For optimal performance, you need more than minimum specifications**

Memor

y

How much RAM do you need?

- **Depends on type of applications you intend to run on your computer**

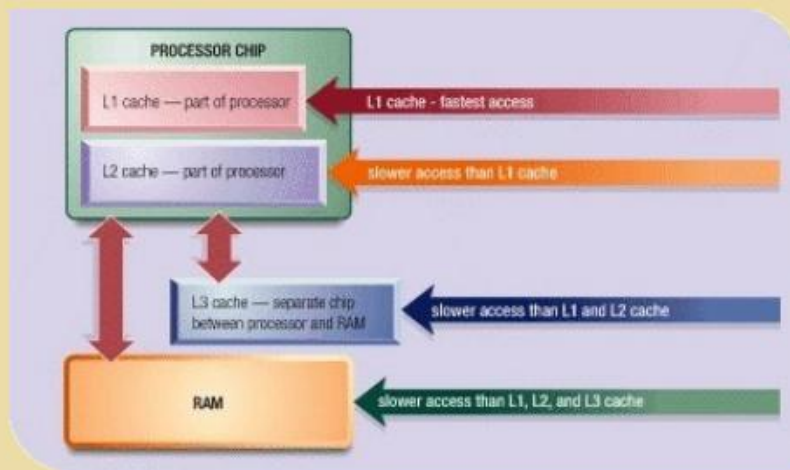
RAM	256 MB to 1 GB	512 MB to 1 GB	2 GB and up
Use	• Home and business users managing personal finance • Using standard application software such as word processing • Using educational or entertainment CD-ROMs • Communicating with others on the Web	• Users requiring more advanced multimedia capabilities • Running number-intensive accounting, financial, or spreadsheet programs • Using voice recognition • Working with videos, music, and digital imaging • Creating Web sites • Participating in video conferences • Playing Internet games	• Power users creating professional Web sites • Running sophisticated CAD, 3D design, or other graphics-intensive software

Memor

y

What is **cache**?

- Helps speed computer processes by storing frequently used instructions and data
- Also called **memory cache**



- L1 cache built into processor
- L2 cache slower but has larger capacity
- L2 advanced transfer cache is faster, built directly on processor chip
- L3 cache is separate from processor chip on motherboard (L3 is only on computers that use L2 advanced transfer cache)

Memor

y

What is **read-only memory (ROM)**?

Memory chips that store permanent data and instructions

Nonvolatile memory, it is not lost when computer's power is turned off

Three types:

Firmware—
Manufactured with permanently written data, instructions, or information

PROM
(programmable read-only memory)—
Blank ROM chip onto which a programmer can write permanently

EEPROM
(electrically erasable programmable read-only memory)—
Type of PROM containing microcode programmer can erase

Memor

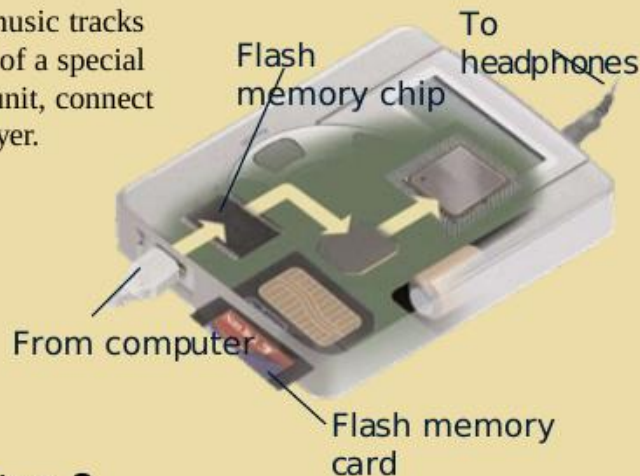
y

What is **flash memory**?

- **Nonvolatile memory that can be erased electronically and rewritten**
- **Used with PDAs, digital cameras, smart phones, music players, digital voice recorders, printers, Internet receivers, and pagers**

Step 1.

Purchase and download MP3 music tracks from a Web site. With one end of a special cable connected to the system unit, connect the other end into the MP3 player.

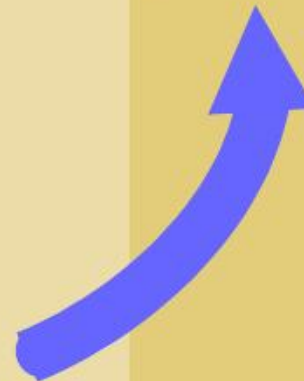


Step 2.

Instruct the computer to copy the MP3 music track to the flash memory chip in the MP3 player.

Step 3.

Plug the headphones into the MP3 player, push a button on the MP3 player, and listen to the music through the headphones.

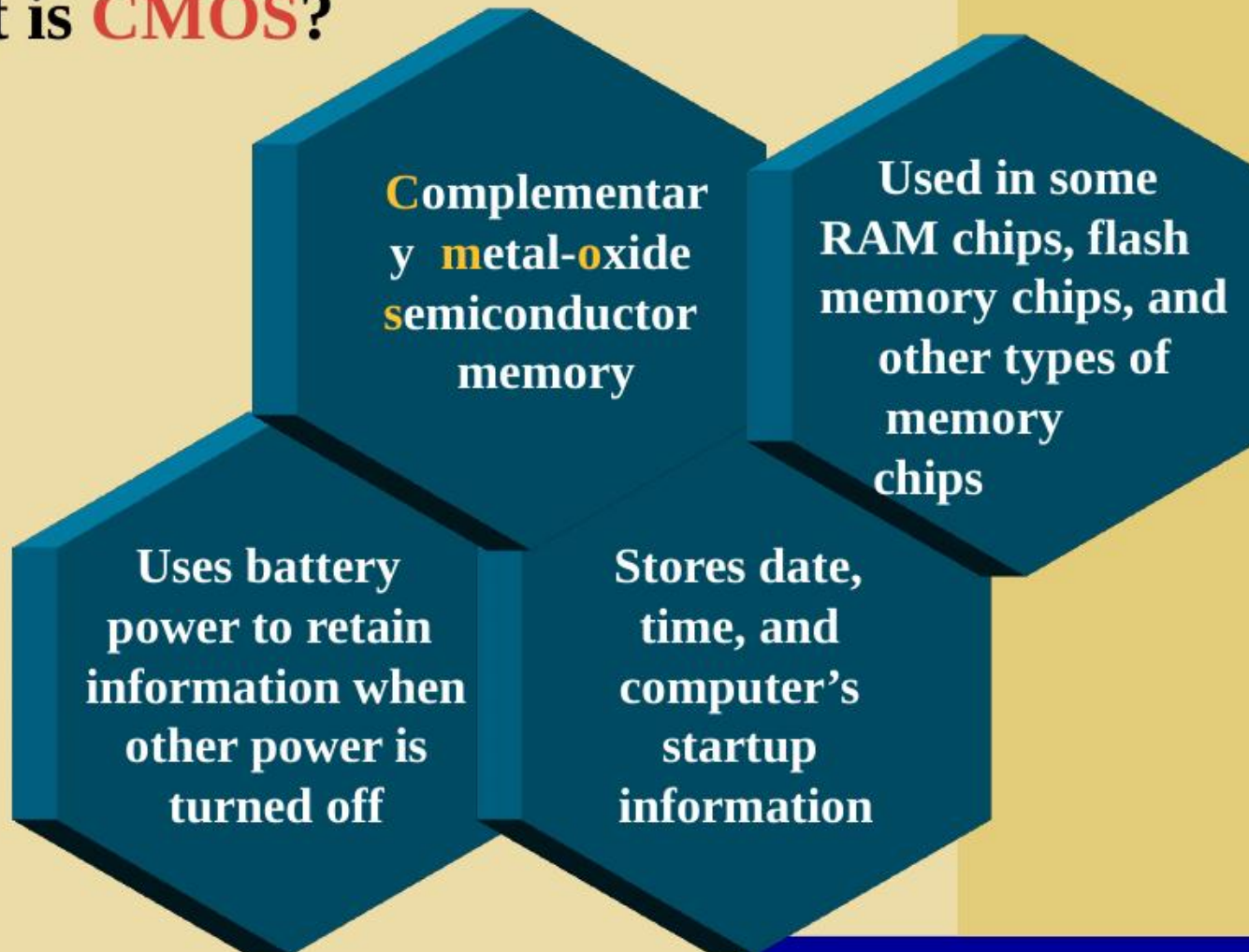


MP3
Player

Memor

y

What is **CMOS**?



Memor

y

What is **access time**?

- Amount of time it takes processor to read data from memory
- Measured in **nanoseconds** (ns), one billionth of a second
- It takes 1/10 of a second to blink your eye; a computer can perform up to 10 million operations in ~~time~~ amount of



Term	Speed
Millisecond	One-thousandth of a second
Microsecond	One-millionth of a second
Nanosecond	One-billionth of a second
Picosecond	One-trillionth of a second

Expansion Slots and Adapter Cards

What is an **adapter card**?

- Enhances system unit or provides connections to external devices called **peripherals**
- Also called an expansion card

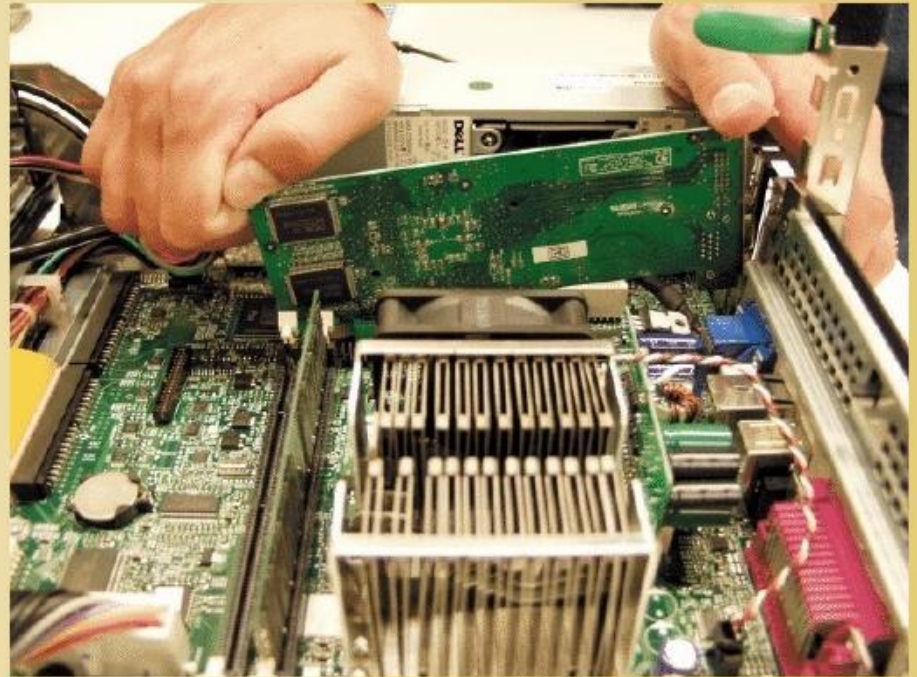
TYPES OF ADAPTER CARDS

Adapter Card	Purpose
Disk controller	Connects disk drives
FireWire	Connects to FireWire devices
Graphics accelerator	Increases the speed at which graphics are displayed
MIDI	Connects musical instruments
Modem	Connects other computers through telephone or cable television lines
Network	Connects other computers and peripherals
PC-to-TV converter	Connects a television
Sound	Connects speakers or a microphone
TV tuner	Allows viewing of television channels on the monitor
USB 2.0	Connects to USB 2.0 devices
Video	Connects a monitor
Video capture	Connects a camcorder

Expansion Slots and Adapter Cards

What is an **expansion slot**?

- An opening, or socket, on the motherboard that can hold an adapter card
- With **Plug and Play**, the computer automatically configures cards and other devices as you install them



Expansion Slots and Adapter Cards

What are **PC cards**, and **flash memory cards**?

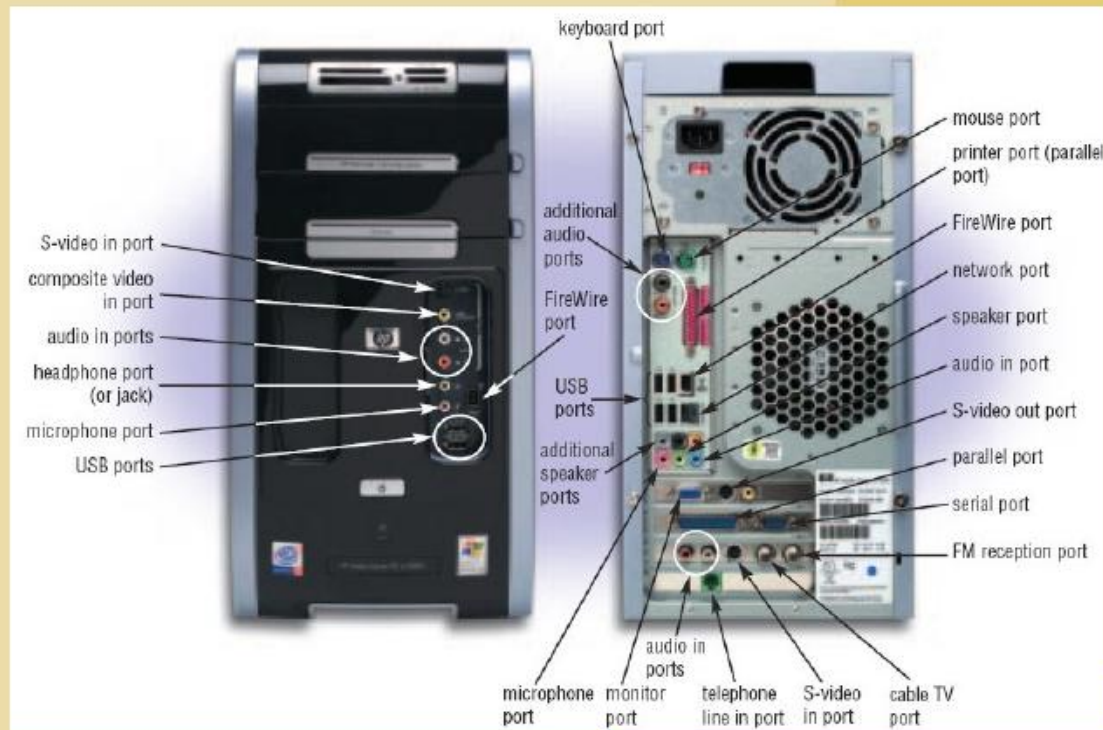
- A **PC card** adds memory, storage, sound, fax/modem, communications, and other capabilities to notebook computers
- A **flash memory card** allows users to transfer data from mobile devices to desktop computers
- USB Flash drive



Ports and Connectors

What are **ports** and **connectors**?

- **Port** connects external devices to system unit
- **Connector** joins cable to peripheral
 - Available in one of two genders: male and female



Ports and Connectors

What are different types of connectors?

CONNECTOR TYPES

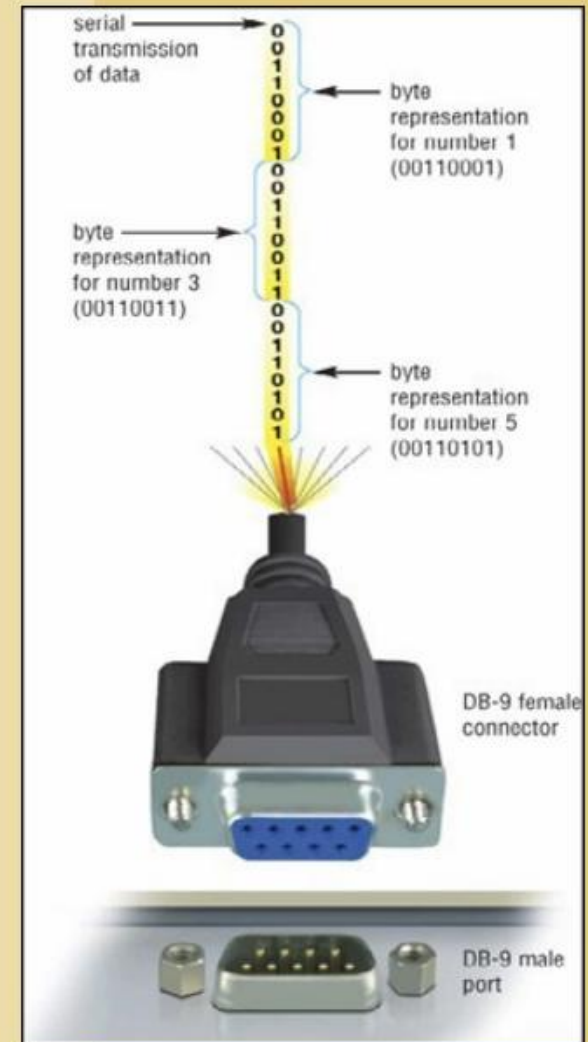
Connector Type	Picture	Connector Type	Picture
Audio in		Mouse	
Cable TV		Network	
Composite video in		Printer	
FireWire		Serial	

FM reception		Speaker	
Headphone		S-video in	
Keyboard		S-video out	
Microphone		Telephone line in	
Monitor		USB	

Ports and Connectors

What is a **serial port**?

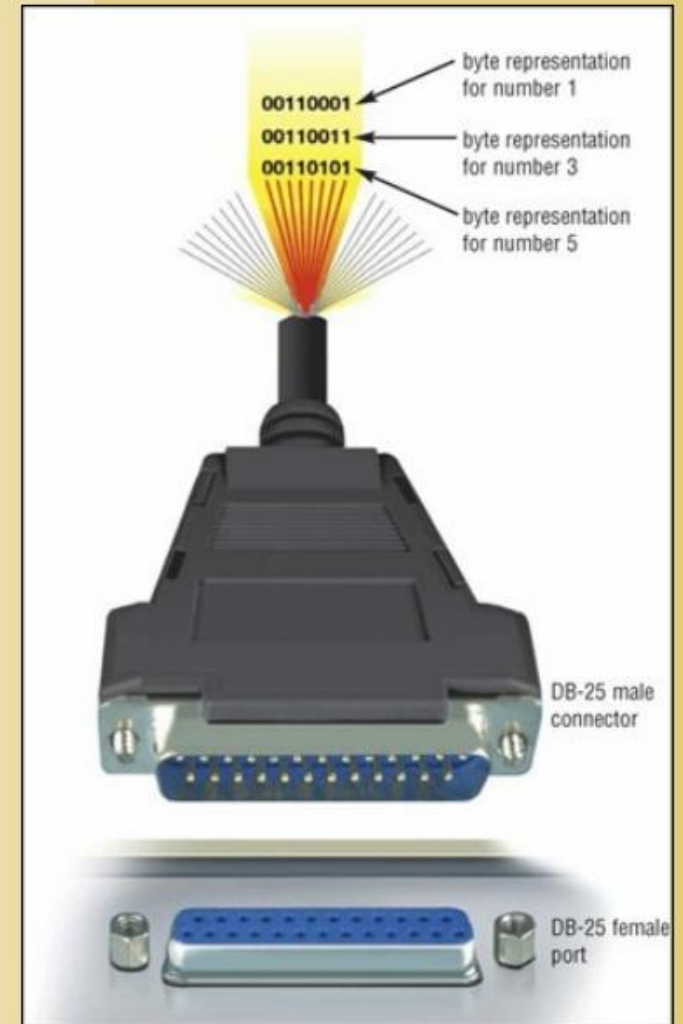
- Transmits one bit of data at a time
- Connects slow-speed devices, such as mouse, keyboard, modem



Ports and Connectors

What is a **parallel port**?

- Connects devices that can transfer more than one bit at a time, such as a printer



Ports and Connectors

What are **USB** ports?

USB (universal serial bus) port can connect up to 127 different peripherals together with a single connector type

PCs typically have six to eight USB ports on front or back of the system unit

Single USB port can be used to attach multiple peripherals in a daisy chain

First USB device connects to USB port on computer

Second USB device connects to first USB device

Third USB device connects to second USB device, and so on



Ports and Connectors

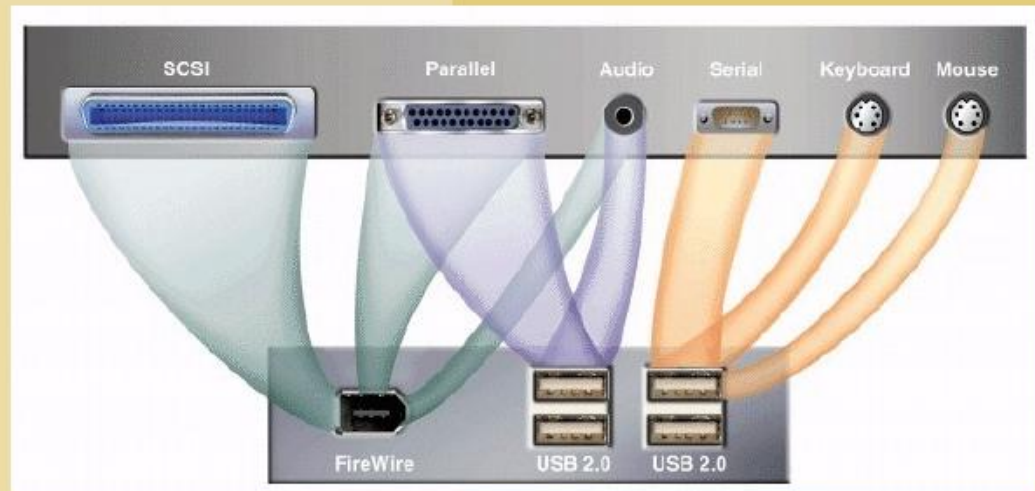
What are **FireWire** ports?

- Connects multiple types of devices that require faster data transmission speeds
- Allows you to connect up to 63 devices together

Ports and Connectors

What are special-purpose ports?

- Allow users to attach specialized peripherals or transmit data to wireless devices
 - MIDI (Musical Instrument Digital Interface) port
 - SCSI (small computer system interface) port
 - IrDA (Infrared Data Association) port
 - Bluetooth port

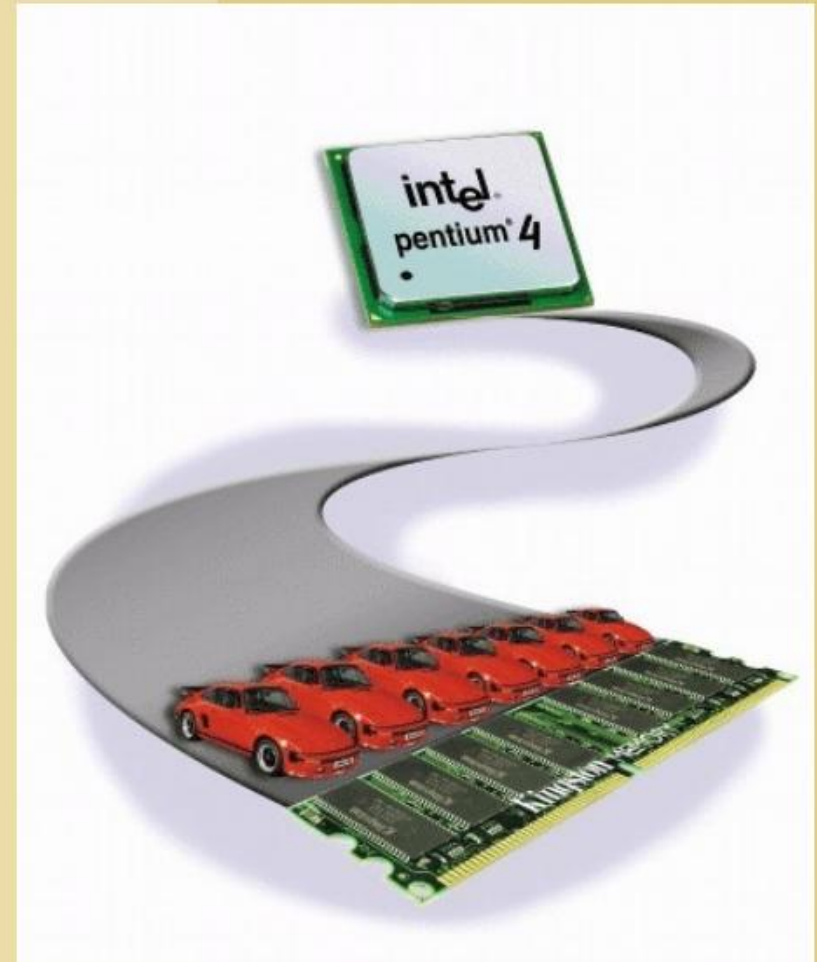


Bus

S

What is a **bus**?

- **Channel that allows devices inside computer to communicate with each other**
 - System bus connects processor and RAM
 - Bus width determines number of bits transmitted at one time
 - **Word size** is the number of bits processor can interpret and execute at a given time



Buse

S

What is an expansion bus?

- **Allows processor to communicate with peripherals**



Bay

S

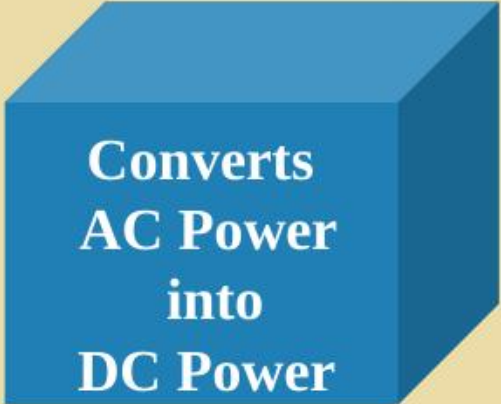
What is a **bay**?

- Open area inside system unit used to install additional equipment
- **Drive bays** typically hold disk drives



Power Supply

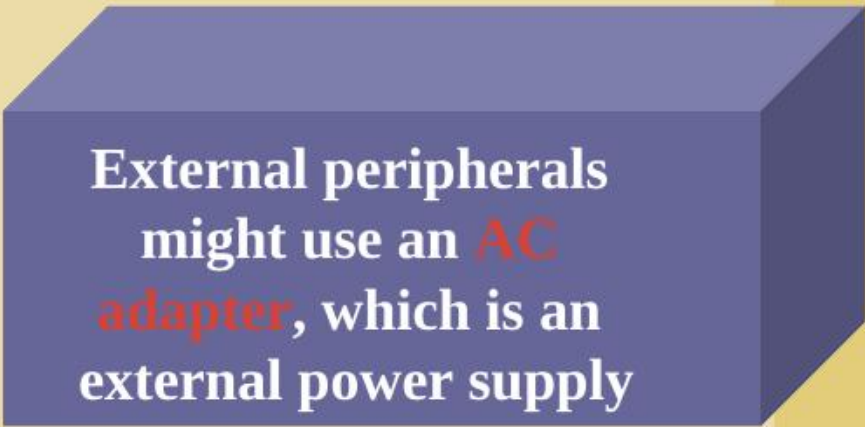
What is a **power supply**?



Converts
AC Power
into
DC Power



Fan keeps
system unit
components
cool

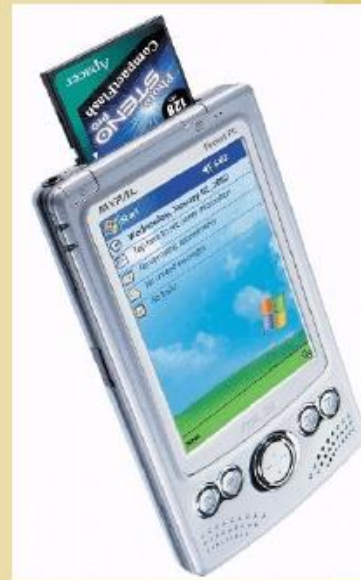
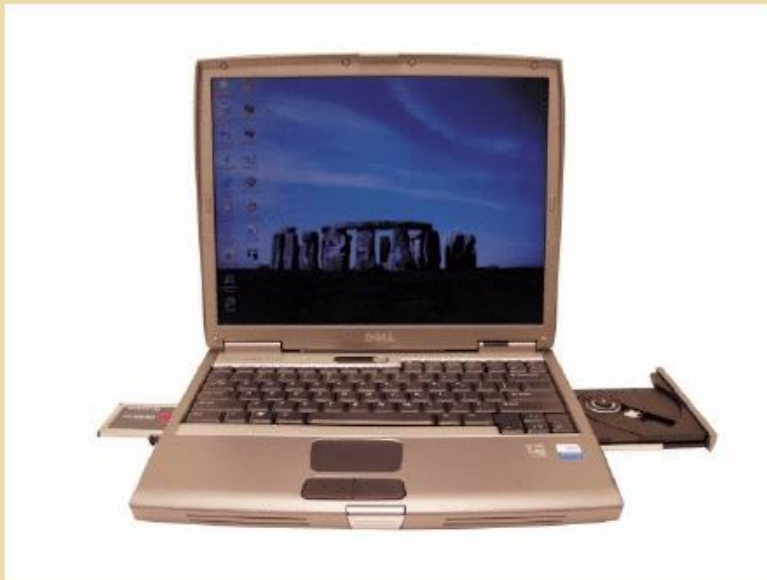


External peripherals
might use an **AC
adapter**, which is an
external power supply

Mobile Computers and Devices

What is a mobile computer?

- Notebook, weighing between 2.5 and 8 pounds, or mobile device such as a PDA



Mobile Computers and Devices

What ports are on a notebook computer?

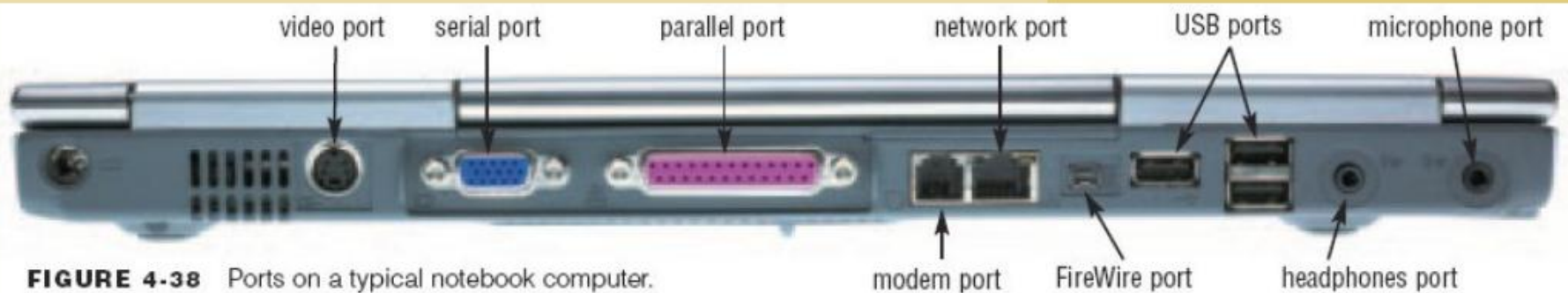
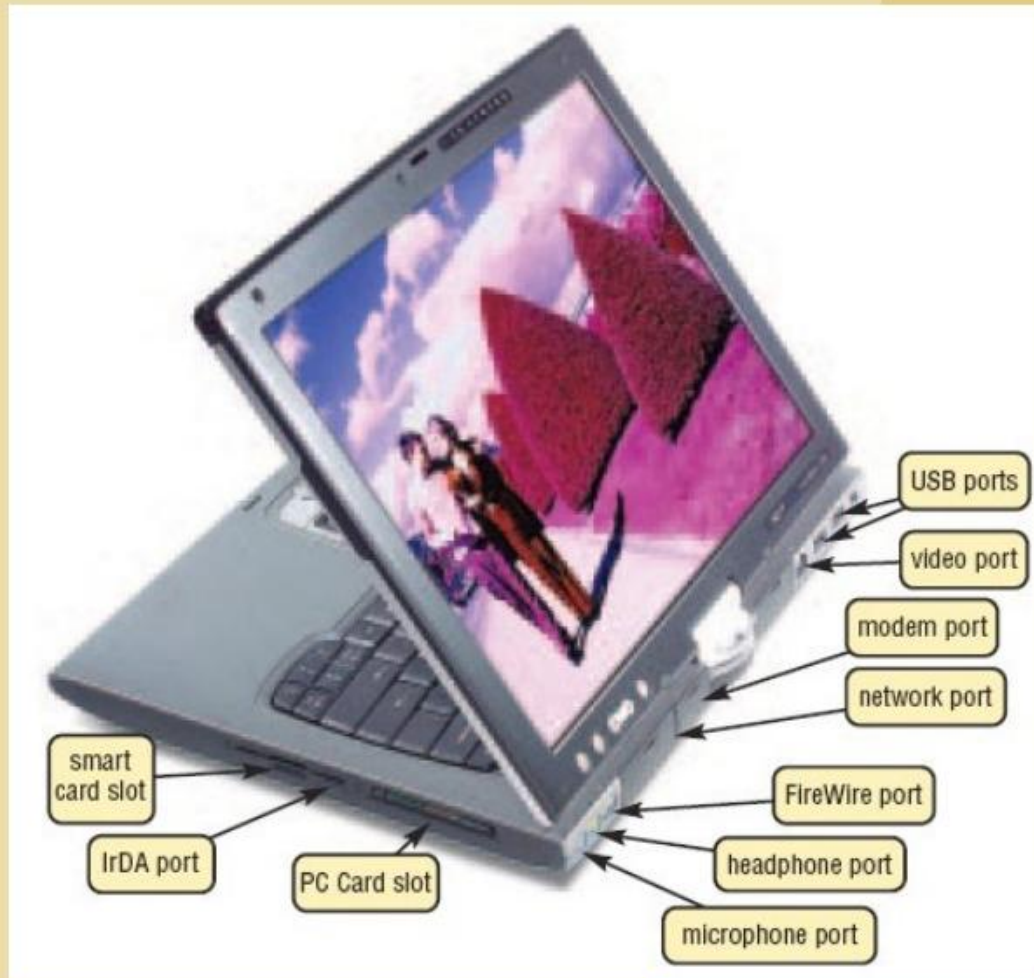


FIGURE 4-38 Ports on a typical notebook computer.

Mobile Computers and Devices

What ports and slots are on a tablet PC?



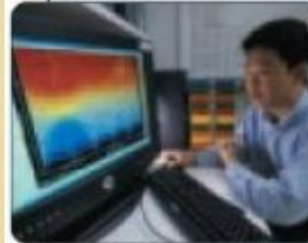
Putting It All Together

What are suggested processor, **clock speed**, and RAM requirements based on the needs of various types of users?

SUGGESTED MINIMUM CONFIGURATIONS BY USER

User	Processor, Clock Speed, and RAM
HOME 	Celeron (3 GHz or higher) or Sempron (2 GHz or higher) or Pentium 4 (2.8 GHz or higher) or Sempron (2 GHz or higher) Minimum RAM: 256 MB
SMALL OFFICE/ HOME OFFICE 	Pentium 4 (3 GHz or higher) or Athlon 64 (2.4 GHz or higher) Minimum RAM: 512 MB
MOBILE 	Celeron M (1.5 GHz or higher) or Pentium M (2 GHz or higher) or Mobile Sempron (1.8 GHz or higher) Minimum RAM: 512 MB

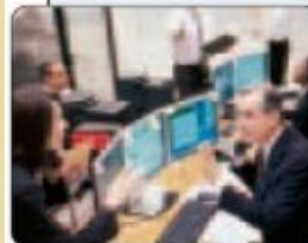
POWER



Itanium 2 (1.6 GHz or higher) or
Opteron (2.4 GHz or higher) or
Xeon MP (3 GHz or higher) or
Athlon MP (2.25 GHz or higher)

Minimum RAM: 2 GB

LARGE BUSINESS



Pentium 4 (3.4 GHz or higher) or
Athlon 64 (2.4 GHz or higher)

Minimum RAM: 1 GB

Keeping Your Computer Clean

Over time, the system unit collects dust – even in a clean environment

- **Preventative maintenance requires a few basic products:**



Clean Your PC

Cleaning out the interior of your PC can increase its life and prevent disaster

